

Igneous intrusions

Some of the most spectacular rock formations on Earth have been created from molten rock forced up through the planet's brittle crust. When it cooled, it crystallized into massive batholiths and plutons, volcanic plugs, walllike dikes, and horizontal sills. As soft surrounding rocks have been worn away, these igneous intrusions are left standing proud in the landscape.



Great Dike

Africa

Location Zimbabwe

Intrusion type Lopolith (T-shaped dike)

Rock Serpentinite

Age 2.6 billion years

This colossal intrusion of ancient, altered mantle rock is 3 miles (5 km) wide and 340 miles (550 km) long, which means that it is only really visible from space. As it formed, the molten rock spread out at the top as a lopolith—a dike that is T-shaped in cross-section.



Devil's Tower

North America

Location Wyoming, USA

Intrusion type Plug or pluton

Rock Phonolite

Age 40 million years

This stupendous monolith of solidified magma has split into hexagonal columns, like the basalt of the Giant's Causeway. Scientists believe that it is either a volcanic plug or a pluton—a mass of igneous rock that formed underground from cooled magma.



Sugar Loaf

South America

Location Rio de Janeiro, Brazil

Intrusion type Batholith

Rock Granite

Age 800 million years

Rising dramatically above Guanabara Bay in Rio de Janeiro, Brazil, the Sugar Loaf is part of an enormous granite batholith

underlying the whole area. The rock formed from magma that erupted between two overlapping oceanic plates.



Great Karoo

Africa



Location South Africa

Intrusion type Dikes and sills

Rock Dolerite

Age 180 million years

In southern Africa, horizontal layers of shale and sandstone are cracked and split by dikes and sills of dolerite, which is like basalt with bigger crystals. The rock erupted during the breakup of an ancient supercontinent called Gondwana.

